Vibrational dependence of the quartic centrifugal constant and the octic centrifugal constant of linear molecules

$-D_{J,vib} = -D_J + \sum_{i=1}^{3N-6} \beta_i \left(v_i + \frac{1}{2}\right)$

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When one solves the anharmonicity problem[1] on the basis of experimental data, one usually employs the observed values of the x constants of the quartic anharmonicity and also the vibrational dependence of the rotational constants α. The number of such measurable quantities is usually limited. In the simple case of a linear triatomic molecule XYZ one can, in general, determine altogether 9 values for the constants x and α while the expansion for the potential energy in terms of the normal coordinates contains 6 cubic coefficients K'_{ijk}, and 9 quartic coefficients k'_{ijkn}. It has been shown in Ref. 2 that the vibrational dependence of the l doubling can also contribute information about K'_{ijk}. In this communication we obtain expressions for the constants β_k, which characterize the dependence of the quartic centrifugal distortion on the vibrational state, and also for the octic centrifugal constant L, which supplies additional information about the cubic and quartic anharmonicity coefficients.

The centrifugal correction to the rotational energy of a linear molecule can be represented in the form[1]

$$E_d = \left[-D_J + \sum_n \beta_n\left(v_n + \frac{1}{2}\right) + \sum_t \beta_t (v_t + 1)\right][J(J+1) - l^2]^2 + H[J(J+1) - l^2]^3 + L[J(J+1) - l^2]^4, \tag{1}$$

where the terms β_n and β_t refer to parallel and perpendicular vibrations, respectively. The terms with β in Eq. (1) are diagonal matrix elements of the operator $\widetilde{H}_{24}$, for which a general expression has been obtained in Ref. 3.

In the case of a linear molecule the general operator expression for $\widetilde{H}_{24}$ can be substantially simplified; after rather tedious calculations one obtains the following expressions for the constants β_n and β_t:

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$$\begin{aligned}\beta_n = {} & 4\omega_n^2 C_n^2\left(\frac{B}{\omega^2} - \frac{D_J}{B^2}\right) + \frac{1}{4}\sum_{n' \neq n''} k'_{nnn'n''} C_{n'} C_{n''} \\ & - \sum_{n'} \frac{k'_{nnn'}}{2\omega_{n'}} r_n - 4B \sum_{n',t} C_{n'}\zeta_{nt}[(3\omega_n^2 + \omega_t^2) \\ & \times \omega_{n'}^{3/2} C_{n'}\zeta_{n't} + (\omega_n^2 + \omega_t^2)\,\omega_n^{3/2} C_n \zeta_{n't}]\,[\omega_n \omega_{n'}^{1/2} (\omega_n^2 - \omega_t^2)]^{-1} \\ & + 4\sum_{t,t'} \{X_{nt}X_{nt'}F^{tt'} + X^{nt}X^{nt'}F_{tt'}\} \\ & - 4\sum_{n',t} \{X_{nt}X^{tn'}F_{n'n} + X^{nt}X_{tn'}F^{n'n}\} \\ & - 16\sum_{n'} \{U_{nn'}^2(\omega_n + \omega_{n'}) + V_{nn'}^2(\omega_{n'} - \omega_n)\},\end{aligned} \tag{2}$$

bend

$$\begin{aligned}\beta_t = {} & \frac{1}{4}\sum_{n,n'} k'_{nn'tt} C_n C_{n'} + B\sum_{n,n'} (\omega_t^2/\omega_n\omega_{n'})^{1/2}\zeta_{nt}\zeta_{n't} - \sum_n k'_{ntt} r_n/2\omega_n \\ & - 4B\sum_{n,n'} C_n \zeta_{tn'}[2\omega_t^2 \omega_{n'}^{3/2} C_{n'}\zeta_{tn} + (3\omega_t^2 + \omega_{n'}^2)\,\omega_n^{3/2} C_n \zeta_{tn'}] \\ & \times [\omega_t \omega_n^{1/2}(\omega_t^2 - \omega_{n'}^2)]^{-1} \\ & + 4\sum_{n,n'} \{X_{tn}X_{tn'}F^{nn'} + X^{tn}X^{tn'}F_{nn'}\} \\ & - 4\sum_{n,t'} \{X_{tn}X^{nt'}F_{t't} + X^{tn}X_{nt'}F^{tt'}\} \\ & - 16\sum_{t'} \{U_{tt'}^2(\omega_t + \omega_{t'}) + V_{tt'}^2(\omega_{t'} - \omega_t)\},\end{aligned} \tag{3}$$

where

$$X^{nt} = 2B\frac{\zeta_{nt}(\omega_n\omega_t)^{1/2}}{\omega_n^2 - \omega_t^2}, \quad X_{nt} = \frac{B\zeta_{nt}}{(\omega_n\omega_t)^{1/2}}\frac{(\omega_n^2 + \omega_t^2)}{(\omega_n^2 - \omega_t^2)}, \tag{4}$$

$$r_n = 4\omega_n C_n\left(\frac{D_J}{B} - \frac{B^2}{\omega_n^2}\right) + \frac{1}{2}\sum_{n',n''} k_{nn'n''} C_{n'} C_{n''}, \tag{5}$$

B is the equilibrium rotational constant ω_n and ω_t are the frequencies of the parallel and perpendicular vibrations, respectively, $\zeta_{nt} = \zeta^x_{n,tb}$ are the Coriolis interaction constants, $C_n = -B_n^{xx}/\omega_n$ where B_n^{xx} are the rotational derivatives ($B_t = 0$), D_J is the equilibrium centrifugal distortion constant, k'_{ijk} and k'_{ijkn} are the third and fourth derivatives of the potential energy of the molecule with respect to the dimensionless normal coordinates[1,3]; and the following notation has been introduced:

$$F^{nn'} = B^2\sum_t \zeta_{nt}\zeta_{n't}(\omega_n\omega_{n'})^{1/2}\frac{(\omega_n^2 + \omega_{n'}^2 - 2\omega_t^2)}{(\omega_n^2 - \omega_t^2)(\omega_{n'}^2 - \omega_t^2)}, \tag{6}$$

$$F_{nn'} = r_{nn'} + B^2\sum_t \frac{\zeta_{nt}\zeta_{n't}}{(\omega_n\omega_{n'})^{1/2}}\frac{(\omega_n^2\omega_{n'}^2 - \omega_t^4)}{(\omega_n^2 - \omega_t^2)(\omega_{n'}^2 - \omega_t^2)}, \tag{7}$$

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$$U_{nn'}=\frac{r_{nn'}}{4(\omega_n+\omega_{n'})}+\frac{B^2}{8}\sum_t\frac{\zeta_{nt}\zeta_{n't}}{\omega_t(\omega_n\omega_{n'})^{1/2}}\frac{(\omega_n+\omega_t)^2(\omega_{n'}-\omega_t)^2+(\omega_n-\omega_t)^2(\omega_{n'}+\omega_t)^2}{(\omega_n^2-\omega_t^2)(\omega_{n'}^2-\omega_t^2)}, \tag{8}$$

$$V_{nn'}=\frac{r_{nn'}}{4(\omega_n-\omega_{n'})}+\frac{B^2}{8}\sum_t\frac{\zeta_{nt}\zeta_{n't}}{\omega_t(\omega_n\omega_{n'})^{1/2}} \tag{9}$$

$$\times\frac{[(\omega_n+\omega_t)^2(\omega_{n'}+\omega_t)^2(\omega_n+\omega_{n'}-2\omega_t)-(\omega_n-\omega_t)^2(\omega_{n'}-\omega_t)^2(\omega_n+\omega_{n'}+2\omega_t)]}{(\omega_n^2-\omega_t^2)(\omega_{n'}^2-\omega_t^2)(\omega_n-\omega_{n'})}, \tag{10}$$

$$r_{nn'}=3\omega_n\omega_{n'}C_nC_{n'}/4B+\frac{1}{2}\sum_{n''}k'_{nn'n''}C_{n''},$$

$$r_{tt'}=\frac{1}{2}\sum_n k_{tt'n}C_n. \tag{11}$$

Expressions for $F^{tt'}$ and $F_{tt'}$ can be obtained from Eq. (6) for $F^{nn'}$ and from Eq. (7) for $F_{nn'}$, respectively, if one substitutes t, t' for the indices, nn', and if one changes the summations over t on the right-hand sides of these equations into summations over n.

The last three terms in Eqs. (2) and (3) can be rearranged and combined. However, the expressions thus obtained cannot be further substantially simplified. All the terms in Eqs. (2) and (3) are written in the same order as the corresponding commutator expressions in Ref. (3). That should facilitate matters for the reader in case he desired to confirm the derivation of Eqs. (2) and (3) starting from the operator expressions of Ref. (3).

The value of the sextic centrifugal constant H in Eq. (1) depends on the cubic anharmonicity coefficients, and the value of the octic centrifugal constant L depends on the cubic and quartic anharmonicities. The formula for H is well known[4,5]; an expression for the constant L in terms of the molecular parameters has been obtained from the general results of Ref. (3). It has the form

quartico completo!

$$L=-\frac{8D_J^3}{B^2}+\frac{1}{24}\sum_{n,n',n'',n'''}k'_{nn'n''n'''}C_nC_{n'}C_{n''}C_{n'''}$$

$$+8BD_J\sum_n\frac{C_n^2}{\omega_n}-\sum_n r_n^2/2\omega_n, \tag{12}$$

where the expression for r_n is given in Eq. (5).

Hence, Eqs. (2), (3), and (12) for the constants β_n, β_t, and L together with the known formulas for q_t^J from Ref. 2 and for H from Refs. (4) and (5) enlarge appreciably the possibilities for the solution of the anharmonicity problem for linear molecules if experimental data for the vibration–rotation constants of high orders are available; this is actually the case for many linear molecules.

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Cylindrical surface electromagnetic waves and the appearance of radial–angular surface structures

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Recently, to explain many phenomena arising at the interface of two media, representations of the excitation of surface electromagnetic modes have been widely adopted. Here two types of such excitations are examined: local modes (surface plasmons), the sizes of localization of which are much less than the wavelength, and nonlocalized [surface electromagnetic waves (SEW)], the path length of which is much greater than the wavelength of the incident radiation λ.[1,2]

Excitation of plasmons occurs at individual inhomogeneities of the surface and at particles located near it.[2] For excitation of SEW, which are usually regarded as plane waves, it is necessary that on the surface there be extended periodic microinhomogeneities of the relief or permittivity ε of the medium (lattice type). At the present time an examination of the properties of SEW is conducted to a considerable degree in connection with the establishment of their important role in the formation of periodic structures during the action on the surface of a number of materials of intense coherent radiation.[3,4] It is considered that such structures are formed on heating and destruction of the surface in the interference field of two plane waves: the incident and a dif-

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